

SUBPROGRAMS IN ABAP (MODULARIZATION TECHNIQUES)

Enhancing code structure with reusable modules

INTRODUCTION TO SUBPROGRAMS

- Subprograms are **modular parts of an ABAP program** used to divide a large program into smaller, manageable units.
- These units are designed mainly for **code reusability**, which means we can write logic once and reuse it many times.
- In real-time SAP projects, large programs are always divided into subprograms to make them **easy to understand, debug, and maintain**.

👉 Main Concept:

SUBPROGRAM = Definition + Calling

TYPES OF SUBPROGRAMS

There are 5 types of subprograms in ABAP:

- Include Programs
- Macros
- Subroutines (FORM routines)
- Function Modules (FM)
- Methods (Object-Oriented ABAP)

👉 All types are used for **modular programming and reusability**.

👉 Mandatory Rules:

- Number Formal and Actual Parameters should be same.
- Properties of both parameters should be same.

INCLUDE PROGRAM

- Include programs are **separate reusable code files**
- They can be included in any program using: "***INCLUDE program_name.***"

Important Points:

- Include programs **do not support parameters.**
- They are used to:
 - Store declarations
 - Write reusable logic
 - Separate large programs into parts
- 👉 Think of INCLUDE as **copy-paste reuse at compile time**

MACROS

- Macros are used for **small repetitive operations**, mainly arithmetic operations
- **Definition:**

DEFINE macro_name.

statements.

END-OF-DEFINITION.

Important Features:

- Maximum **9 parameters only** → (&1 to &9)
- Used only inside the program
- Cannot be reused in another program
- Fast execution but **difficult to debug**

SUBROUTINES

- Subroutines are **most commonly used subprograms**
- Syntax: *FORM subroutine_name.*
 statements.
 ENDFORM.
- Calling: *PERFORM subroutine_name*

Important Point:

- 👉 **Call first, define later**

Parameter Passing:

- TABLES → Internal tables
- USING → Input variables
- CHANGING → Output variables

LIMITATIONS:

- Cannot handle exceptions
- Cannot be called across servers

FUNCTION MODULES

Function Modules are **advanced subprograms**

Features:

- Global subprograms
- Can be used in multiple programs
- Can be called across servers
- Support exception handling

👉 Created using **SE37 (Function Builder)**

PARAMETERS IN FUNCTION MODULE

- IMPORTING → input variables
- EXPORTING → output variables
- TABLES → internal tables
- CHANGING → input + output
- EXCEPTIONS → error handling

EXAMPLE OF FUNCTION MODULE

```
REPORT ZDEMO_DLOAD NO STANDARD PAGE HEADING.
```

```
data it type table of kna1.
```

```
select * from kna1 into table it up to 10 rows.
```

```
CALL FUNCTION 'GUI_DOWNLOAD'  
  EXPORTING  
    FILENAME          = 'D:\CUSTOMERS.TXT'  
    APPEND             = 'X'  
    WRITE_FIELD_SEPARATOR = 'X'  
  TABLES  
    DATA_TAB         = IT  
  .  
IF SY-SUBRC EQ 0.  
  WRITE :/ 'FILE DOWNLOADED SUCCESSFULLY'.  
ENDIF.
```

FINAL COMPARISON: *INCLUDE* vs *SUBROUTINE* vs *FUNCTION MODULE*

Feature	Include Programs	Subroutines (FORM)	Function Modules (FM)
Definition	Reusable code file included in program	Local subprogram inside a program	Global subprogram stored in function group
Purpose	Code reuse without parameters	Modularization of logic within program	Reusable logic across multiple programs
Syntax	INCLUDE prog_name.	FORM...ENDFORM	CALL FUNCTION
Calling Method	Automatically included (no call required)	PERFORM statement	CALL FUNCTION statement
Parameter Passing	✗ Not allowed	✓ USING, CHANGING, TABLES	✓ IMPORTING, EXPORTING, CHANGING, TABLES
Scope	Can be used in multiple programs	Mostly local (can call externally)	Global (accessible everywhere)
Reusability Level	Medium	Medium	High

FINAL COMPARISON: *INCLUDE* vs *SUBROUTINE* vs *FUNCTION MODULE*

Exception Handling	✗ Not supported	✗ Not supported	✓ Supported (EXCEPTIONS)
Server Access	✗ No	✗ No	✓ Yes (RFC possible)
Execution Type	Code is directly inserted at compile time	Called during runtime	Called during runtime
Debugging	Difficult (mixed with main program)	Easy	Easy (separate object)
Storage Location	Separate include file	Same program	Function Group (SE37)
Real-time Usage	Declarations, screens, common code	Small logic, helper functions	Business logic, standard SAP functions

THANK YOU

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